

Dual-Band Concentric Split Square Resonators Based Metamaterial Absorber for X-Band Applications

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ABSTRACT This manuscript presents a simple and novel design of a dual-band with square split resonators based metamaterial absorber (DBSSRMMA). The two absorptance bands at 8.2 GHz and 10.87 GHz merged to a broad-band that covers entire X-band (8 GHz – 12 GHz). The DBSSRMMA is fabricated on double sided FR4 copper plate with electrical dimensions of $0.194\lambda_0 \times 0.194\lambda_0 \times 0.065\lambda_0$. The unit cell of metamaterial absorber consists of concentric square resonators with opposite splits inclined at 45° . The novelty lies in the response and design of the proposed structure that exhibits dual-band and broad-band without use of parasitic elements, or multilayered structure, or extended substrate width. The response of DBSSRMMA structure is stable with variation in incidence and polarization angle. The metamaterial behaviour is studied in terms of effective values of normalized matched impedance, permeability and permittivity. The proposed DBSSRMMA exhibits reflectance of -24.29 dB at 8.2 GHz and -30.2 dB at 10.87 GHz with peak to peak average absorptance of 97.25. The first absorptance peak is contributed by outer split square resonator and the second absorptance peak is contributed by inner split square resonator. The two bands (at 8.2 GHz and 10.87 GHz) merged to generate a broad-band of -3 dB absorptance bandwidth of 6.21 GHz and -10 dB reflectance bandwidth of 4.41 GHz. The proposed DBSSRMMA structure exhibited polarization and wide incidence angle insensitive response that finds applications in stealth technology, RCS reduction and 5G spectrum absorption.

KEYWORDS: absorber, absorptance, dual-band, broad-band

I. INTRODUCTION

Artificially engineered materials known as “metamaterials” have extraordinary capabilities, like the capacity to regulate and modify the behavior of incoming EM waves [1]. Due to their unique properties, including negative permittivity, permeability, and refractive index [2], as well as their small size and weight [3], these materials have recently drawn more attention. These characteristics make metamaterials useful in a variety of applications, including “sensing” [4], “detecting” [5], “shielding” [6], “absorbing” [7], and “PV-based energy harvesting” [8]. Since the development of the first ideal metamaterial absorber in 2008 [9], there has been a lot of interest in the use of metamaterial in absorber design, especially in microwave and optical frequency applications. Metamaterial absorbers are becoming more and more popular because of their excellent performance, low profile, and flawless absorption capabilities [10]. Additionally, these absorbers can be used in terahertz [11] and infrared [12] applications. In recent years, the research on multi-band and broad-band metamaterial absorbers is more as compared to single-band metamaterial absorbers. In this regard, the absorber's performance is the primary consideration. The multi-band (“dual-band”, “triple-band”, “quad-band”) and broad-band response generated by use of parasitic elements [13], or multi-layered substrate structure

[14], or enhancing the substrate width [15]. Wave shielding from Wi-Fi frequencies at 2.4 GHz and 5 GHz, using a dual band metamaterial absorber based on a symmetrical resonator loaded with metallic stubs [16]. The angular and polarization stable maximum absorptance of 99.2% and 99.9% at 2.4 GHz and 5 GHz is achieved by the MMA unit cell fabricated on a $20 \times 20 \times 1.6 \text{ mm}^3$ FR4 substrate. Dual-band absorption at 1.8 GHz and 3.5 GHz using a metamaterial absorber has been proposed in [17] with distinctive resonant patch design made out of two altered circular rings with angular stability for both incidence angle and polarization angle fluctuation, designed for EMI protection against 5G and GSM networks. Ag nanowire screen-printed second-order cross-fractal structures were used to propose a dual-band metamaterial absorber with optical transparency in [18]. Two inductive-capacitive (L-C) resonances at 6.45 and 21.14 GHz are used to generate the dual-band absorption property with potential for C- and K-band radar stealth applications. The research in [19] proposed a dual-band metamaterial microwave absorber that operates at 2.5 GHz and 5.8 GHz. the unit cell of the absorber is made up of a ring and a circular patch with slit resonator structures. With absorptance greater than 97%, the proposed absorber is useful in absorbing and sensing applications with wide-angle receptivity and polarization insensitive behavior. A polarization and wide angular stable dual-band MMA with absorptance peaks at 6.78 GHz

(91.73%) and 13.92 GHz (99.97%) consists of folded arms square split ring resonator (FAS-SRR) based unit cell is proposed in [20]. In [21], a dual-band MMA consists of two metallic ring resonators, suitable for fifth-generation (5G) applications with absorptance peaks at 28 GHz and 38 GHz with high absorption rates of 98 and 98.4%, respectively with polarization and angular stable is proposed. In [22], a novel metamaterial-based 5G absorber with dual bands is introduced, which is made up of symmetric meander lines connected by a transmission line and operates at 24 GHz and 28 GHz. Metamaterial absorber with dual bands for solid and liquid sensing applications is reported in [23]. Dual absorption peaks are exhibited by the absorber at 3.470 GHz for the S-band and 7.219 GHz for the C-band with absorptance greater than 98%.

In the reported designs of dual-band metamaterial absorbers the design is complex, there may or may not be angular and polarization stability, large dimensions and lack of equivalent circuit model. So keeping in mind these limitations a novel design of dual-band metamaterial absorber is proposed with very simple and small scaled design with angular and polarization stability and the absorber exhibits broad-band absorptance (6.21 GHz) with merge of the two bands (at 8.2 GHz and 10.87 GHz) with applications in stealth technology and absorption in 5G spectrum.

II. DESIGN STRUCTURE OF DBSSRMMA UNIT CELL

As seen in Figure 1, the suggested DBSSRMMA unit cell is composed of a middle layer as substrate, top layer as resonating patches, and bottom layer as metallic ground plane. The substrate is FR4 with 2.4 mm thickness with dielectric constant of 4.3 and a loss tangent of 0.02. The patches on the top layer are artificially engineered so that under perfectly impedance matched conditions the incident EM waves are minimally reflected and maximally absorbed at targeted frequencies. The incident EM waves pass through the substrate and prevented to get transmitted by the presence of the metallic ground plane. The so reflected EM waves get attenuated inside the substrate after multiple reflections. The overall dimensions of the unit cell are 7.1 mm x 7.1 mm x 2.4 mm ($0.194\lambda_0$ x $0.194\lambda_0$ x $0.065\lambda_0$), λ_0 at 8.2 GHz. The frequency selective surface (FSS) consists of two square shaped patches with splits on the opposite sides.

The optimized dimensions of the unit cell are depicted in Table 1. Due to its rotational symmetry, the suggested overall structure serves as the perfect absorber. To optimize the dimensions, a number of numerical simulations were performed with the "CST Microwave Studio Suite. Parametric analysis is done whereby varying one parameter and keeping all other parameters constant. The optimization process is done in stages as shown in Figure 2. In stage-1, the simple square patches of different dimensions are taken. In stage-2, the splits are carved out on the two opposite sides (top and bottom) of each square. In stage-3, the outer square patch is rotated by $+45^\circ$ and Inner square is rotated by -45° . In stage-4, the two square patches are brought

together one inside the other to form the final optimised structure of the unit cell.

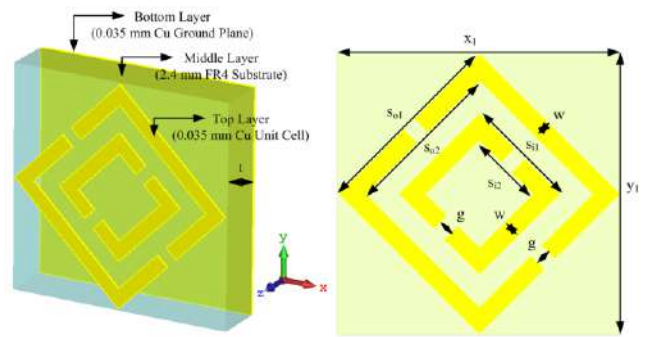


FIGURE 1. Dimensional structure of DBSSRMMA unit cell.

TABLE I. Units for Magnetic Properties

Parameter	Dimension (mm)
x_1	7.1
y_1	7.1
S_{01}	4.94
S_{11}	2.82
w	0.5
g	0.4
t	2.4

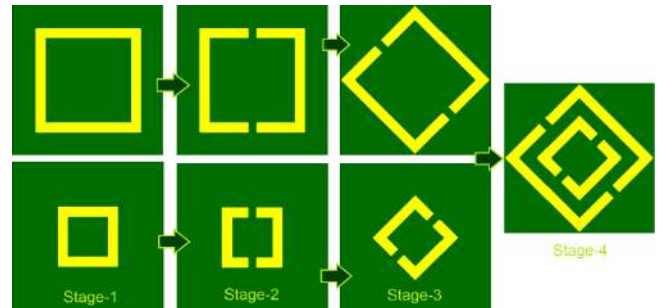


FIGURE 2: Stage wise optimization of the DBSSRMMA unit cell.

III. EQUIVALENT CIRCUIT MODEL OF DBSSRMMA UNIT CELL

The transmission line theory is used to create the MMA's equivalent circuit, where each metallic strip can be modeled as an inductor. On the other hand, the gap between the two metallic conductors, is modeled as a capacitor. In the proposed DBSSRMMA unit cell structure, the FR4 substrate separates the top layer concentric square split resonators from the copper backplane. Consequently, the outer split square copper ring resonator and the copper backplane is modeled as an L_1C_1 circuit. A comparable L_2C_2 circuit is also modeled for the inner split square copper ring resonator and the copper backplane as shown in Figure 3. Both capacitive and inductive effects are also present in the mutual coupling between the two concentric split square ring resonators. Furthermore, there are parasitic capacitances between the MMA and the ports. Since the substrate is the component that contributes to the absorption of the signal, it can be

modeled as the electrical circuit's energy-consuming resistor. Taking all of these factors into account, Figure 4 shows the model of an equivalent circuit

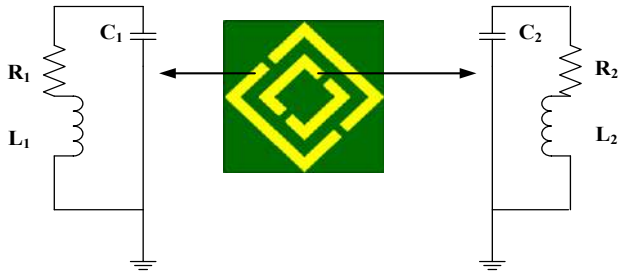


FIGURE 3: Equivalent circuit model of individual ring resonators of the DBSSRMMA unit cell.

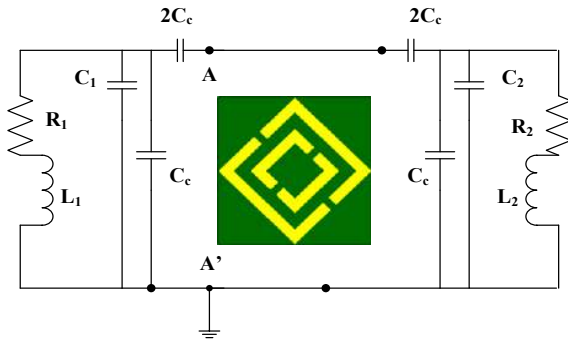


FIGURE 4: Equivalent circuit model of the DBSSRMMA unit cell.

IV. CST SIMULATION OF DBSSRMMA UNIT CELL

The suggested DBSSRMMA unit cell is simulated using the “CST microwave studio suite”. This electromagnetic computational software is used to simulate microwave devices like antennas, metamaterial unit cell, waveguide and other microwave components. It is used to design the proposed DBSSRMMA unit cell, and a frequency domain solver is used to assess its performance. This solver uses numerical analysis to solve electromagnetic field equations using the finite element method (FEM). It splits the device under test into tetrahedral or hexahedral meshes and computes the interaction of these meshes with EM waves using maxwell's equations. Depending on the chosen excitation and the applied boundary conditions, CST calculates the fields around the material by taking resonances, reflections, and refractions into account. The simulation setup is shown in Figure 5. The proposed absorber's unit cell is positioned between two waveguide ports, with the ground plane at port 2 and the resonating patch at port 1's side. The wave propagates in the Z-direction. For the typical incident wave, the boundary condition accounts for magnetic and electrical boundary conditions in the Y and X-axis directions resp. is implemented during the simulation process. Floquet boundary conditions are used that offer flexibility in adjusting the incidence wave angle and polarization angle to study the impact of polarized signals. This port system applies unit cell boundaries in the X and Y directions.

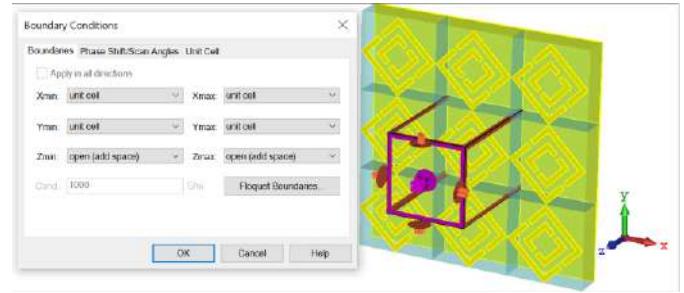


FIGURE 5: Boundary conditions applied to the DBSSRMMA unit cell.

V. SIMULATION RESULTS OF DBSSRMMA STRUCTURE

The proposed DBSSRMMA structure is excited by the wave coming from port 1, and near-zero reflection results electromagnetic resonance at 8.2 GHz and 10.87 GHz. The ground copper layer obstructs the propagation of the transmitted signal after it passes through the substrate layer. Its thickness exceeds the depth of the skin, thus there is no transmission. The intended frequency signal is thus trapped inside the substrate, undergoes a number of internal reflections, and is eventually absorbed in the substrate layer. The absorptance $A(f)$ is calculated using Equation (1) [24], where S_{11}^2 is the reflectance and S_{21}^2 is the transmittance. The ground plane stops the transmission of EM waves through the absorber so the transmittance is reduced to zero and absorptance is solely a function of the reflectance given by Equation (2).

$$A(f) = 1 - S_{11}^2 - S_{21}^2 \quad (1)$$

$$A(f) = 1 - S_{11}^2 \quad (2)$$

Under normalized matched impedance conditions the simulated results in Figure 6 shows the reflection coefficients of -25.62 dB at 8.2 GHz and -31.63 dB at 10.87 GHz for normal incidence of EM waves. The absorptance is estimated using Equation (2) and the Figure 7 depicts the absorptance of 99.7% at 8.2 GHz and 99.9% at 10.87 GHz. The average peak to peak absorptance from 8.2 GHz to 10.87 GHz is 97.25%. The results show high absorptance at the two bands within X-band.

At 8.2 GHz and 10.87 GHz, the surface current density is calculated for the suggested structure in order to create a clear grasp of the absorption mechanism. When time-varying electromagnetic waves normally strike the suggested DBSSRMMA structure parallel to the axis of the concentric square split resonator patches, surface currents are generated. This depends on the MMA resonant characteristics at various resonant frequencies, as illustrated in Figure 8. The color of the surface current indicates its intensity, while the arrow indicates its direction. A magnetic field produced by induced currents either reinforces or counteracts the incident electromagnetic field. Every square ring functions as a resonator, resonating at a certain frequency where absorption takes place. The unit cell's periodic array creates strong coupling between

resonator structures, leading to high absorption of electromagnetic waves. The composite MMA structure results in unique features, including negative permeability

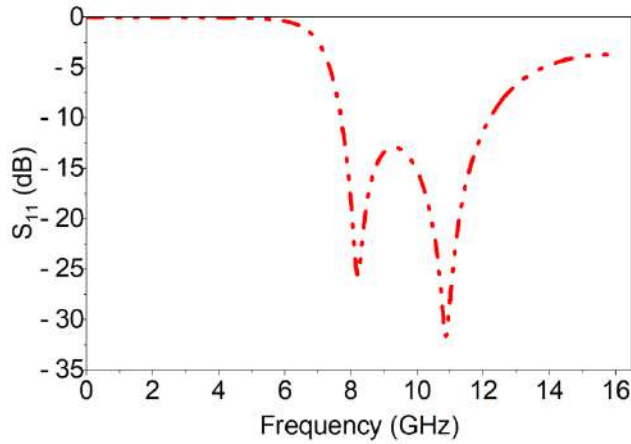


FIGURE 6: Reflectance $S_{11}(f)$ of the DBSSRMMMA structure.

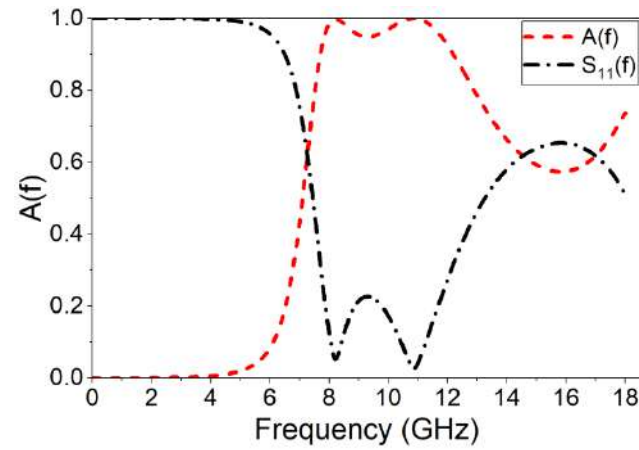


FIGURE 7: Linear Absorbance $A(f)$ and Reflectance $S_{11}(f)$.

TABLE II: Simulated results of the DBSSRMMMA unit cell.

Frequency (GHz)	$S_{11}(f)$ (dB)	$A(f)$ (%)
8.2	-25.62	99.7
10.87	-31.63	99.9

and permittivity. Figure 8 clearly depicts that for first resonance frequency at 8.2 GHz, the surface current is distributed more densely on the outer square resonator as compared to the inner square resonator that contributes to the absorption of first absorbance peak. At 10.87 GHz, the surface current is distributed both on outer and inner square resonators that indicates the combined responsibility of absorption of both resonators. The individual ring resonators contribute to significant absorption at specific resonance peaks, while the combined impact of the ring resonators leads to prolonged broad-band absorption of incident EM waves. The 8.2 GHz absorbance band merged with 10.87 GHz absorbance band to generate a broad-band

of 6.21 GHz (-3 dB absorbance bandwidth) and broad-band of 4.41 GHz (-10 dB reflectance bandwidth).

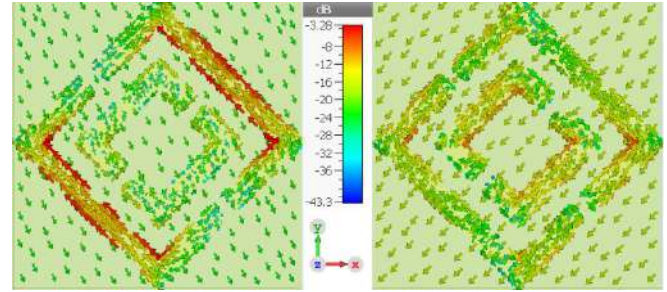


FIGURE 8: Surface current distribution at (a) 8.2 GHz. (b) 10.87 GHz.

VI. METAMATERIAL BEHAVIOUR OF DBSSRMMMA STRUCTURE

For a perfect MMA, two conditions must be met. In order for the incident EM wave to be entirely transferred into the MMA, it must be precisely impedance matched. Furthermore, a high imaginary value of the permeability and permittivity is necessary for the absorbed EM wave to be highly dissipated within the MMA [25]. Figure 9 confirms impedance matching at maximal absorbance peaks, as $\text{Re}[Z_{\text{eff}}(f)]$ approaches 1 and $\text{Im}[Z_{\text{eff}}(f)]$ approaches 0. The normalized impedance matched values are estimated using Equation (3).

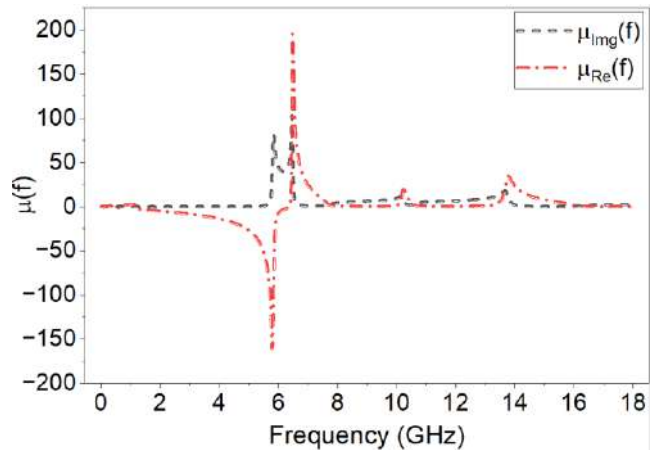


FIGURE 9: Real and Imaginary impedance values of DBSSRMMMA structure.

To study the metamaterial behavior of the proposed MM absorber, dispersion characteristics of simulated S-parameters are calculated using conventional methods such as the “Nicolson Rose Weir (NRW) method” and the “Transmission-Reflection (TR) method” [26]. These methods require S_{11} and S_{21} parameters to calculate effective impedance; $Z_{\text{eff}}(f)$, effective permittivity; $\epsilon_{\text{eff}}(f)$ and effective permeability; $\mu_{\text{eff}}(f)$. The MMA structural design influences effective permittivity and permeability [given by Equation (6) and Equation (7)], which contribute to its effectiveness as a medium. Numerical simulation optimizes the unit cell to obtain MMA values of ϵ_2 and μ_2 , ensuring signal loss within and values of ϵ_1 and μ_1 , ensuring signal propagation within MMA. The Lorentzian transitions

of $\text{Re}[\epsilon_{\text{eff}}(f)]$ and $\text{Re}[\mu_{\text{eff}}(f)]$ and high positive or negative values of $\text{Im}[\epsilon_{\text{eff}}(f)]$ and $\text{Im}[\mu_{\text{eff}}(f)]$ confirms the metamaterial behavior of MMA as depicted in Figure 10.

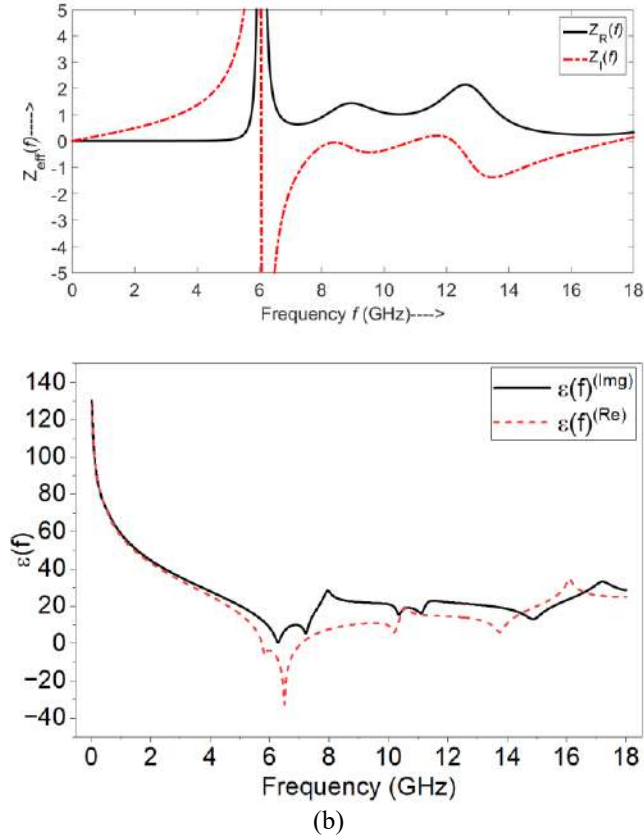


FIGURE 10: Effective values of (a) Permeability $\mu_{\text{eff}}(f)$ (b) Permittivity $\epsilon_{\text{eff}}(f)$ of DBSSRMMA structure.

$$Z_{\text{eff}}(f) = \sqrt{\frac{[1+S_{11}(f)]^2 - S_{21}^2(f)}{[1-S_{11}(f)]^2 - S_{21}^2(f)}} \quad (3)$$

$$\mu_{\text{eff}}(f) = \frac{2}{jk_0 d} \left(\frac{1+S_{11}-S_{21}}{1-S_{11}+S_{21}} \right) \quad (4)$$

$$\epsilon_{\text{eff}}(f) = \frac{2}{jk_0 d} \left(\frac{1-S_{11}-S_{21}}{1+S_{11}+S_{21}} \right) \quad (5)$$

$$\mu_{\text{eff}}(f) = \mu_1(f) + j\mu_2(f) \quad (6)$$

$$\epsilon_{\text{eff}}(f) = \epsilon_1(f) + j\epsilon_2(f) \quad (7)$$

VII. EXPERIMENTAL SETUP AND MEASURED RESULTS

There are two possible methods of measuring: one waveguide port measurement method and other far-field measurement method. The two horn antennas used in the far-field method are placed next to each other. One operates as the generator of the EM wave, while the other receives the reflected wave. In our case the same antenna acts as a transmitter and a receiver as depicted in Figure 11. The device is placed at far field distant. Using this method, the measurement is made inside an anechoic chamber that is confined by the foam based conventional absorbers, which

eliminates undesired reflected signals. The measurement is recorded in two steps. In step one, the ground plane facing the antenna is placed at the same height as that of the antenna. The horn antenna transmits the EM waves and receives the reflected wave and the measured reflectance is recorded in VNA connected to the horn antenna. In step two, the top layer with concentric square split patches faces the horn antenna and VNA records the reflectance. The difference of the two measured values gives the calibrated reflectance. This process is implemented to measure reflectance for normal incidence as well as with variation in θ (incidence angle) and ϕ (polarization angle).

The proposed DBSSRMMA structure fabricated on 2.4 mm double sided FR4 substrate by photolithography method followed by wet etching process. The fabricated MMA consists of 28 rows and 28 columns of 784 unit cells with overall dimensions of $198.8 \times 198.8 \times 2.4 \text{ mm}^3$ as shown in Figure 12. Implementing the two step measurement process as described above for normal incidence of the EM waves the reflectance measured to be -24.29 dB at 8.2 GHz and -30.2 dB at 10.87 GHz with peak to peak average absorptance of 97.25 as compared to simulated absorptance of -25.62 dB at 8.2 GHz and -31.63 dB at 10.87 GHz as depicted in Figure 13. The measured values are in close resemblance with measured values. Nevertheless, an incident wave in practical applications won't necessarily be incident normal for the MMA. A perfect MMA should have same response to EM waves coming at any angle. The angular stability is therefore measured with variation in angle of incidence (θ) and in polarization angle (ϕ). The measured setup is depicted in



FIGURE 11: Experiment setup (anechoic chamber).

TABLE III: Simulated verses measured results of the DBSSRMMA structure.

Frequency (GHz)	Simulated $S_{11}(f)$ (dB)	Measured $S_{11}(f)$ (dB)	Simulated $A(f)$ (%)	Measured $A(f)$ (%)
8.2	-25.62	-24.29	99.72	99.62
10.87	-31.63	-30.20	99.99	99.90

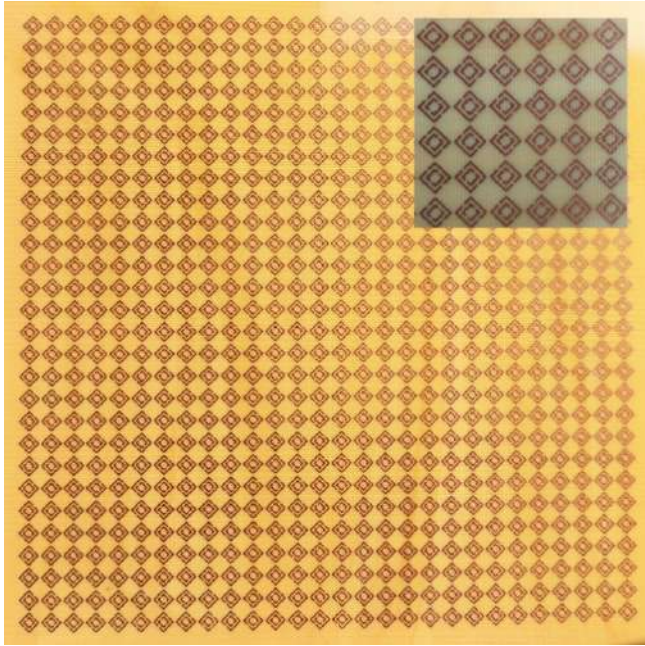


FIGURE 12: Fabricated DBSSRMMA structure.

Figure 14(a) and 15(a) resp. Figure 14(b) shows the proposed DBSSRMMA structure response with variation of polarization angle ($\varphi = 0^\circ$, $\varphi = 90^\circ$, $\varphi = 180^\circ$). The response is identical and the results conclude the polarization insensitivity of proposed DBSSRMMA structure.

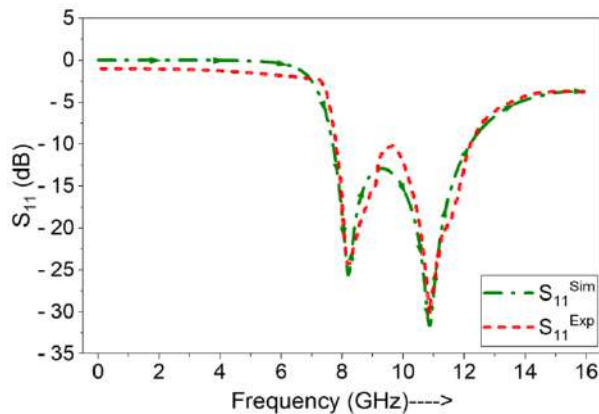


FIGURE 13: Comparison of simulated and measured reflectance.

The response of MMA is also measured with variation in angle of incidence ($\theta = 10^\circ$, $\theta = 20^\circ$, $\theta = 30^\circ$) as depicted in Figure 15 (b). The results show that as the incident angle is increased the proposed DBSSRMMA structure's response is affected beyond 30° , with lesser availability of the surface to the incident EM waves. The results conclude the partial insensitive response to the variation of incident angle.

VIII. CONCLUSION

A simple and novel design of MMA is simulated, fabricated and tested. High absorptance is achieved by artificially engineering the concentric square patches with splits on opposite sides with unit cell dimensions of 7.1 mm x 7.1

mm x 2.4 mm and fabricated MMA with overall dimensions of 19.88 cm x 19.88 cm x 0.24 cm. The MMA is fabricated on double sided FR4 substrate. The proposed DBSSRMMA structure exhibited absorptance at two frequency bands; 8.2 GHz and 10.87 GHz with reflectance (absorptance) of -24.29 dB (99.62%) and -30.20 dB (99.90%) resp. Surface current distribution is analyzed to study the absorption behavior and contribution of individual resonating patches towards the absorptance bands. The analysis revealed that outer square patch contributes towards first absorptance band and inner square patch for second absorptance band. The combined effect of the resonating patches results in the broad-band of -3dB absorptance bandwidth of 6.21 GHz and -10 dB reflectance bandwidth of 4.41 GHz. Effective values of normalized matched impedance, permeability and permittivity are analyzed to study the metamaterial behaviour. The response of the proposed DBSSRMMA structure is found to stable at normal incidence of EM waves as well with variation in

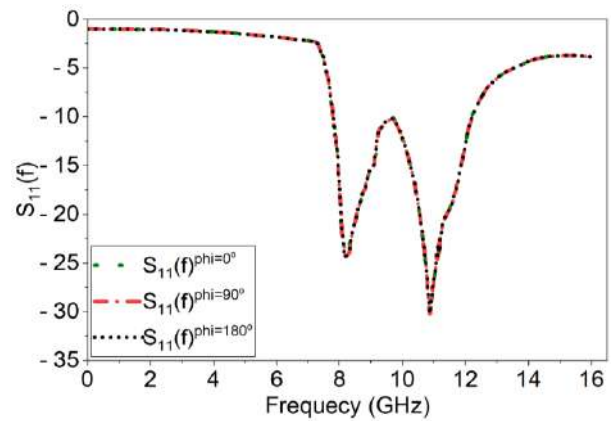


FIGURE 14: Reflectance measurement with variation in polarization angle (φ).

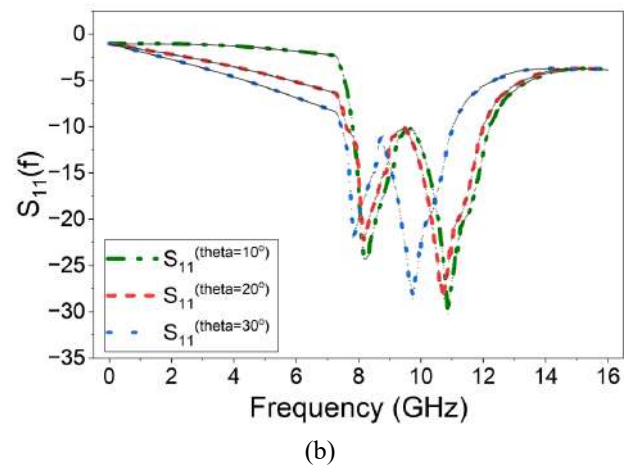


FIGURE 15: Reflectance measurement with variation in incidence angle (θ).

polarization angle (φ) and angle of incidence (θ). The proposed DBSSRMMA structure is compared with recently reported (2023-2025) MMA designs in terms of unit cell design, design complexity, unit cell dimensions, dual-band absorptance, angular stability, polarization stability, broad-

band and equivalent circuit model. The Table 5 comparison clearly depicts the proposed DBSSRMMA structure outperforms the reported designs in terms of design simplicity, high and stable response with broad-band absorptance in X-band (8 GHz – 12 GHz) that finds application in stealth technology, RCS reduction and 5G spectrum absorption.

REFERENCES

- [1] J. B. Pendry, D. Schurig and D. R. Smith, "Controlling electromagnetic fields," *Science*, 312 (2006): 1780-1782, <https://doi.org/10.1126/science.1125907>.
- [2] V. G. Veselago, "The electrodynamics of substances with simultaneously negative values of ϵ and μ ," *Soviet Physics Uspekhi*, 10, no. 4, (1968): 509-514, <https://iopscience.iop.org/article/10.1070/PU1968v010n04A/BEH003699>.
- [3] K. Errajaji, N. Jebbor, S. Das, T. Islam, B. T. P. Madhav, T. El-Arrouch, "Design and analysis of a multi-band miniaturized metamaterial absorber for wireless communication applications," *Optical and Quantum Electronics*, 56,(2024):232, <https://doi.org/10.1007/s11082-023-05813-6>.
- [4] A. Q. Alsulami, S. Wageh, Ahmed A. Al-Ghamdi, R. M. H. Bilal, and M. A. Saeed, "A Tunable and Wearable Dual-Band Metamaterial Absorber Based on Polyethylene Terephthalate (PET) Substrate for Sensing Applications" *Polymers*, 14, no. 21, (2022): 4503, <https://doi.org/10.3390/polym14214503>.
- [5] M. R. Rakhshani, F. Kazemi, and M. Rashki, "Metamaterial absorber using cascaded ring resonators and optimization through machine learning for sensing applications", *Scientific Reports*, 15, (2025): 39977, <https://doi.org/10.1038/s41598-025-23594-8>.
- [6] A.A.M. Rahman, M. T. Islam, Md. Moniruzzaman, H. Alsaif, A.A. Maash, M. M. Hasan, and M. S. Soliman, "Resonator-based near perfect metamaterial absorber with high EMI shielding for Wi-Fi and 5G applications", *International Journal of Optomechatronics*, 18, no. 1, (2024): <https://doi.org/10.1080/15599612.2024.2375497>.
- [7] A. Agrawal and M. Misra, "Angle of Incidence and Polarization Insensitive Ultra-Thin Ultra Wide-Band Metamaterial Absorber based on Novel Non-Linearity in Unit Cell Design", *Journal of Microwaves, Optoelectronics and Electromagnetic Applications*, 20, no. 2, (2021): 556-569, <http://dx.doi.org/10.1590/2179-10742021v20i31207>.
- [8] Y. Gai, S. Zhou, and G. Lan, "Ultra-broadband Perfect Metamaterial Absorber for Solar Energy Harvesting", *Plasmonics*, 20, (2025): 5145–5153, <https://doi.org/10.1007/s11468-024-02711-9>.
- [9] N. I. Landy, S. Sajuyigbe, J. J. Mock, D. R. Smith, and W. J. Padilla, "Perfect metamaterial absorber," *Physical Review Letters*, 100, (2008): 207402-1 - 207402-4, <https://doi.org/10.1103/PhysRevLett.100.207402>.
- [10] K. H. M. Naik, A. K. Singh, and D. R. Krishna, "Ultrathin wideband low profile circuit analog metamaterial absorber", *Microwave and Optical Technology Letters*, 66, no. 7, (2024): 1-8, <https://doi.org/10.1002/mop.34247>.
- [11] S. S. Pahadsingh, and B. Appasani, "Metamaterial-Based Terahertz Absorbers for Refractive Index Sensing: Types, Mechanism, and Applications", *Plasmonics*, 20, (2025): 5557–5571, <https://doi.org/10.1007/s11468-024-02532-w>.
- [12] Z. Cao, X. Zeng, and G. Lan, "Ultra-broadband metamaterial absorber for near-infrared and mid-infrared applications optimized via deep learning", *Applied Optics*, 64,(2025):7527-7533, <https://doi.org/10.1364/AO.573376>.
- [13] Z. Özer, V. Akdoğan, L. Wang, and M. Karaaslan, "Graphene- Based Tunable Metamaterial Absorber for Terahertz Sensing Applications", *Plasmonics*, 19,(2024):3381–3391, <https://doi.org/10.1007/s11468-024-02249-w>.
- [14] D. Zhang, Y. Chen, and Q. Zhou, "Multilayer broadband metamaterial for microwave absorption based on resistive frequency selective surface", *Results in Physics*, 73, (2025): 1-8, <https://doi.org/10.1016/j.rinp.2025.108261>.
- [15] J. Zhu, Y. Sheng, L. Chen, G. Gao, M. Shi, Z. Yin, and J. Yang, "W-Band Ultra-Thin Broadband Metamaterial Absorber—Design and Applications", *Photonics*, 12, (2025): 1-21, <https://doi.org/10.3390/photonics12030282>.
- [16] M. Moniruzzaman, M. T. Islam, M. F. Mansor, M. S. Soliman, N. Misran, and M. Samsuzzaman, "Tuning metallic stub loaded symmetrical resonator based dual band metamaterial absorber for wave shielding from Wi-Fi frequencies", *Alexandria Engineering Journal*, 63,(2023):399-414, <https://doi.org/10.1016/j.aej.2022.07.051>.
- [17] M. Moniruzzaman, S. Larguech, M. Mobarak, N. M. Jizat, S. S. Alharbi, M. T. Islam, M. Samsuzzaman, and S. S. Al-Bawri, "Dual band polarization insensitive metamaterial absorber for EMI shielding from GSM and 5G communication systems", *Scientific Reports*, 15, (2025):12292, <https://doi.org/10.1038/s41598-025-96507-4>.
- [18] S. Bark, K. Junghyeon, L. Minjae, and L. Sungjoon "Optically Transparent Dual-Band Metamaterial Absorber Using Ag Nanowire Screen-Printed Second-Order Cross-Fractal Structures", *Fractal and Fractional*, 8,no.3,(2024):153, <https://doi.org/10.3390/fractalfract8030153>.
- [19] M. S. Sim, K. Y. You, R. Dewan, F. Esa, M. R. Salim, S. Y. N. Kew, F. Hamid, "Dual-Band Metamaterial Microwave Absorber using Ring and Circular Patch with Slits", *Advanced Electromagnetics*, 12, no. 4, (2023): 36-44, <https://doi.org/10.7716/aem.v12i4.2324>.
- [20] M. Berka, U. Özkaya, T. Islam, M. El-Ghzaoui, S. Varakumari, S. Das, and Z. Mahdjaub "A miniaturized folded square split ring resonator cell based dual band polarization insensitive metamaterial absorber for C- and Ku-band applications", *Optical Quantum Electronics*, 55, (2023): 699, <https://doi.org/10.1007/s11082-023-04954-y>.
- [21] M. Edries, H. A. Mohamed, and A. A. Ibrahim, "Dual Band 28/38 GHz Metamaterial Absorber for 5G Applications", *Journal of Infrared, Millimeter, and Terahertz Waves*, 44, (2023): 898–911, <https://doi.org/10.1007/s10762-023-00948-9>.
- [22] S. A. Naqvi, M. A. Baqir, G. Gourley, A. Iftikhar, M. Saeed Khan, and D. E. Anagnostou, "A Novel Meander Line Metamaterial Absorber Operating at 24 GHz and 28 GHz for the 5G Applications", *Sensors*, 22, no. 10,(2022): 3764, <https://doi.org/10.3390/s22103764>.
- [23] M.A. Khalil, W. H. Yong, M. S. Islam, L. Y. Chiong, A. Hoque, N. Ullah, H. H. GOH, T. A. Kurniawan, M. S.

- Soliman, and M. T. Islam, "Design of dual peak star shaped metamaterial absorber for S and C band sensing applications", *Scientific Reports*, 14, (2024): 26609, <https://doi.org/10.1038/s41598-024-77215-x>
- [24] A. Agrawal, M. Misra, and A. Singh "Oblique Incidence and Polarization Insensitive Multiband Metamaterial Absorber with Quad Paired Concentric Continuous Ring Resonators," *Progress In Electromagnetics Research M*, 60 (2017): 33–46, <https://www.jpier.org/PIERM/pier.php?paper=17061302>.
- [25] A. Agrawal, M. Misra, and A. Singh, "Wide incidence angle and polarization insensitive dual broad-band metamaterial absorber based on concentric split and continuous rings resonator structure", *Material Research Express (IOP Science)*, 5, no. 11, (2018): 115801, <https://iopscience.iop.org/article/10.1088/2053-1591/aadd83>.